

BE IT Second Year 2nd Semester Class Test (I) Examination 22-23
Subject: Object Oriented Systems

Full Marks — 30

Time: 1 Hour

General Instructions:

1. This question paper consists of 2 questions, all are compulsory. However internal choice has been provided on some questions. Read these carefully before attempting any questions.
2. Write your name and roll number on every sheet.
3. Different parts of the same question should be answered together.
4. All symbols have their usual meaning unless otherwise stated.
5. Assume suitable data, if necessary.
6. Candidates are requested to take the test honestly and ethically.
7. If any Identical answer found then all of them are penalized (Marks will be deducted). Remember, copying is cheating.

CO1: Differentiate different object oriented programming language and **Solve** problems by developing Java programs using (i) classes, (ii) inheritance, (iii) nested classes and (iv) Exceptions (K3) **(15 Marks)**

- 1) Answer any **three** of the following:
 - a) Define abstraction in OOS. Explain it with suitable example. Differentiate different OOP languages like C++, Java and Python. **[1+2+3]²**
 - b) Show with a complete program how static methods can be overloaded. Invoke each of the overloaded methods from the main() method. Show how a static member variable of a class can be accessed outside that class definition. **[3+2]**
 - c) Consider an abstract class having an abstract method and a constructor. Show (with suitable code) how this abstract class can be inherited and also show the way of invoking its constructor and the abstract method. **[3+2]**
 - d) Discuss the significance of each word in public static void main (String args[]). **[5]**

CO2: Solve problems using thread programming and Input-Output (K3) **(15 Marks)**

- 2) Answer a) and any **one** from remaining:
 - a) What is Suppose a developer wrote a class A
 Then wrote another class B that extend class A
 Now Inside the body of the class B developer want to use java thread.
 - i. Is it possible to do so? Justify your answer.
 - ii. Your next task is to help the developer by writing the appropriate code snippet. **[(1+2)+5]**
 - b) Discuss the functionality of sleep() method along with its various overloaded versions. How can you invoke the run() method of a thread? How can you invoke the run() method of a thread? **[3+2+2]**
4+3
 - c) Categorized java IO. Explain all of them using suitable code. **[7]**